

高等微积分

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第六章：求导的逆运算—不定积分





普通高等教育“十五”国家级规划教材

数学分析教程

(上册)

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小结: 不定积分

- 将定义在区间上的函数 f 的原函数的一般表达式称为 f 的不定积分, 记作 $\int f(x) dx$. 这是一个以 x 为自变量的函数.
- 若 $f \in \mathcal{C}[a, b]$, 则 $\int f(x) dx = \int_a^x f(t) dt + C$.

不定积分与导数、微分的关系

• 若 $\int f(x) dx = F(x) + C$, 则 $F'(x) = f(x)$,

$$\left(\int f(x) dx \right)' = F'(x) = f(x),$$

$$dF(x) = f(x) dx, \quad d\left(\int f(x) dx \right) = f(x) dx,$$

$$\int f(x) dx = \int F'(x) dx = \int dF(x) = F(x) + C.$$

- (线性性) $\forall \alpha, \beta \in \mathbb{R}$, 我们有

$$\int (\alpha f(x) + \beta g(x)) \, dx = \alpha \int f(x) \, dx + \beta \int g(x) \, dx.$$

基本的不定积分公式

- $\int dx = x + C.$
- $\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + C \ (\alpha \neq -1),$
- $\int \frac{1}{x} dx = \ln |x| + C.$

- $\int a^x dx = \frac{a^x}{\ln a} + C \quad (a > 0, \ a \neq 1),$
 $\int e^x dx = e^x + C.$
- $\int \sin x dx = -\cos x + C, \int \cos x dx = \sin x + C.$

- $\int \operatorname{sh} x \, dx = \operatorname{ch} x + C, \int \operatorname{ch} x \, dx = \operatorname{sh} x + C.$
- $\int \sec^2 x \, dx = \tan x + C.$
- $\int \csc^2 x \, dx = -\cot x + C.$
- $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C.$

- $\int \frac{dx}{1+x^2} = \arctan x + C.$
- $\int \frac{dx}{\sqrt{x^2+a^2}} = \ln |x + \sqrt{x^2 + a^2}| + C.$
- $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln |x + \sqrt{x^2 - a^2}| + C.$

回顾：求不定积分的基本方法

- 第一换元积分法 (凑微分): 若 $F'(y) = f(y)$, 则

$$\int f(u(x))u'(x)dx = \int f(u(x))du(x) = F(u(x)) + C.$$

回顾: 求不定积分的基本方法

- 第二换元积分法: 如果 $f(x(t))x'(t) = F'(t)$, 则

$$\begin{aligned}\int f(x) dx &\stackrel{x=x(t)}{=} \int f(x(t))x'(t) dt \\ &\stackrel{t=t(x)}{=} F(t(x)) + C.\end{aligned}$$

- $\int \sec x \, dx = \ln |\sec x + \tan x| + C.$
- $\int \csc x \, dx = \ln |\csc x - \cot x| + C.$

回顾: 求不定积分的基本方法

下面假设 $a > 0$.

- 若含 $\sqrt{a^2 - x^2}$, 作变换 $x = a \sin t$ ($|t| \leq \frac{\pi}{2}$).
- 若含 $\sqrt{x^2 + a^2}$, 作变换 $x = a \tan t$ ($|t| < \frac{\pi}{2}$).
- 若含 $\sqrt{x^2 - a^2}$, 要分情况讨论: 当 $x > a$ 时, 定义 $x = a \sec t$ ($0 \leq t < \frac{\pi}{2}$); 而当 $x < -a$ 时, 定义 $x = -u$ 或 $x = -a \sec t$ ($0 \leq t < \frac{\pi}{2}$).

§ 6.2. 方法三: 分部积分法

设函数 u, v 均为一阶连续可导. 由于

$$d(uv) = u dv + v du,$$

则 $u dv = d(uv) - v du$, 于是

$$\int u dv = uv - \int v du.$$

例 26. 计算 $\int x \cos x \, dx$.

解:
$$\begin{aligned} \int x \cos x \, dx &= \int x \, d(\sin x) \\ &= x \sin x - \int \sin x \, dx = x \sin x + \cos x + C. \end{aligned}$$

例 27. 计算 $\int \ln x \, dx$.

解:
$$\begin{aligned}\int \ln x \, dx &= x \ln x - \int x d(\ln x) \\ &= x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - x + C.\end{aligned}$$

例 28. 计算 $\int \arcsin x \, dx$.

解:
$$\begin{aligned}\int \arcsin x \, dx &= x \arcsin x - \int x \, d(\arcsin x) \\&= x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} \, dx = x \arcsin x - \int \frac{d(x^2)}{2\sqrt{1-x^2}} \\&= x \arcsin x + \int \frac{d(1-x^2)}{2\sqrt{1-x^2}} = x \arcsin x + \int d\sqrt{1-x^2} \\&= x \arcsin x + \sqrt{1-x^2} + C.\end{aligned}$$

例 29. 计算 $\int xe^x dx$.

解: $\int xe^x dx = \int x d(e^x) = xe^x - \int e^x dx = xe^x - e^x + C$.

例 30. 计算 $\int 3x^2 \arctan x dx$.

$$\begin{aligned}\text{解: } \int 3x^2 \arctan x dx &= \int \arctan x d(x^3) \\ &= x^3 \arctan x - \int x^3 d(\arctan x) \\ &= x^3 \arctan x - \int \frac{x^3}{1+x^2} dx \\ &= x^3 \arctan x - \int \left(x - \frac{x}{1+x^2} \right) dx \\ &= x^3 \arctan x - \frac{1}{2}x^2 + \frac{1}{2} \int \frac{d(1+x^2)}{1+x^2} \\ &= x^3 \arctan x - \frac{1}{2}x^2 + \frac{1}{2} \ln(1+x^2) + C.\end{aligned}$$

例 31. 计算 $\int x \ln^2 x \, dx$.

解:
$$\begin{aligned} \int x \ln^2 x \, dx &= \int \ln^2 x \, d\left(\frac{x^2}{2}\right) \\ &= \frac{x^2}{2} \ln^2 x - \int \frac{x^2}{2} \, d(\ln^2 x) \\ &= \frac{1}{2} x^2 \ln^2 x - \int \frac{x^2}{2} \cdot \frac{2 \ln x}{x} \, dx \\ &= \frac{1}{2} x^2 \ln^2 x - \int x \ln x \, dx \\ &= \frac{1}{2} x^2 \ln^2 x - \int \ln x \, d\left(\frac{x^2}{2}\right) \\ &= \frac{1}{2} x^2 \ln^2 x - \frac{1}{2} x^2 \ln x + \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx \\ &= \frac{1}{2} x^2 \ln^2 x - \frac{1}{2} x^2 \ln x + \frac{x^2}{4} + C. \end{aligned}$$

例 32. 计算 $\int \sqrt{a^2 - x^2} dx$ ($a > 0$).

解:
$$\begin{aligned}\int \sqrt{a^2 - x^2} dx &= x\sqrt{a^2 - x^2} - \int x d(\sqrt{a^2 - x^2}) \\&= x\sqrt{a^2 - x^2} + \int \frac{x^2}{\sqrt{a^2 - x^2}} dx \\&= x\sqrt{a^2 - x^2} + \int \left(\frac{a^2}{\sqrt{a^2 - x^2}} - \sqrt{a^2 - x^2} \right) dx \\&= x\sqrt{a^2 - x^2} + a^2 \int \frac{d(\frac{x}{a})}{\sqrt{1 - (\frac{x}{a})^2}} - \int \sqrt{a^2 - x^2} dx \\&= x\sqrt{a^2 - x^2} + a^2 \arcsin \frac{x}{a} - \int \sqrt{a^2 - x^2} dx.\end{aligned}$$

由此我们立刻可得

$$\int \sqrt{a^2 - x^2} dx = \frac{1}{2} \left(x\sqrt{a^2 - x^2} + a^2 \arcsin \frac{x}{a} \right) + C.$$

例 33. 计算 $\int \sqrt{x^2 + a^2} \, dx \, (a > 0)$.

$$\begin{aligned}\text{解: } \int \sqrt{x^2 + a^2} \, dx &= x\sqrt{x^2 + a^2} - \int \frac{x^2 \, dx}{\sqrt{x^2 + a^2}} \\ &= x\sqrt{x^2 + a^2} - \int \left(\sqrt{x^2 + a^2} - \frac{a^2}{\sqrt{x^2 + a^2}} \right) \, dx \\ &= x\sqrt{x^2 + a^2} + a^2 \ln |x + \sqrt{x^2 + a^2}| - \int \sqrt{x^2 + a^2} \, dx,\end{aligned}$$

由此立刻可得

$$\int \sqrt{x^2 + a^2} \, dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2} \ln |x + \sqrt{x^2 + a^2}| + C.$$

例 34. 设 $m \in \mathbb{N}^*$. 求 $I_m = \int \frac{dx}{(x^2+a^2)^m}$ ($a > 0$).

解:
$$\begin{aligned} I_m &= \frac{x}{(x^2+a^2)^m} - \int x d\frac{1}{(x^2+a^2)^m} \\ &= \frac{x}{(x^2+a^2)^m} + 2m \int \frac{x^2 dx}{(x^2+a^2)^{m+1}} \\ &= \frac{x}{(x^2+a^2)^m} + 2m \int \frac{dx}{(x^2+a^2)^m} - 2ma^2 \int \frac{dx}{(x^2+a^2)^{m+1}} \\ &= \frac{x}{(x^2+a^2)^m} + 2mI_m - 2ma^2 I_{m+1}. \end{aligned}$$

于是 $I_{m+1} = \frac{x}{2a^2m(x^2+a^2)^m} + \frac{2m-1}{2a^2m} I_m$. 注意到

$$I_1 = \int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan \frac{x}{a} + C,$$

由此可得 I_m 的一般表达式.

例 35. 计算 $\int e^{ax} \cos bx dx$, $\int e^{ax} \sin bx dx$ ($ab \neq 0$).

解: 利用分部积分可得

$$\begin{aligned}\int e^{ax} \cos bx dx &= \int e^{ax} d\left(\frac{1}{b} \sin bx\right) \\ &= \frac{1}{b} e^{ax} \sin bx - \frac{a}{b} \int e^{ax} \sin bx dx, \\ \int e^{ax} \sin bx dx &= \int e^{ax} d\left(-\frac{1}{b} \cos bx\right) \\ &= -\frac{1}{b} e^{ax} \cos bx + \frac{a}{b} \int e^{ax} \cos bx dx,\end{aligned}$$

由此立刻可得

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C,$$



同学们辛苦了!